

Benchmarking Protein–Protein Interface Prediction on Experimental versus AlphaFold-Predicted Structures

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Abstract

PeSTo is a geometric deep-learning model that predicts protein–protein interface residues from atomic coordinates, trained only on experimentally determined structures. Because most protein structures available today are AlphaFold predictions, we ask whether PeSTo’s accuracy transfers to AlphaFold input. We assembled 566 protein chains from PDB complexes released after AlphaFold2’s training cutoff, filtered to under 30% sequence identity to PeSTo’s own training, validation, and test sequences, then ran PeSTo on each chain’s isolated experimental structure and on the matched AlphaFold model trimmed to the same residue range. Across 554 chains with defined metrics, PeSTo’s per-chain area under the precision–recall curve (AUPR) was higher on experimental input by a median of 0.027 (95% CI [0.019, 0.035], Wilcoxon $p = 3.5 \times 10^{-17}$), consistent in direction across eight strata and under an alternative buried-surface-area interface definition. The gap was far from uniform: within AlphaFold’s own confidence bands, the pooled AUPR gap shrank from 0.254 in the lowest-confidence residues to 0.039 in the highest-confidence residues, and a per-residue regression associated both AlphaFold confidence (pLDDT) and local structural divergence with each residue’s prediction shift, independently of one another and of solvent exposure. Chain length was not associated with the size of the drop. These results describe a modest but consistent, confidence-linked accuracy cost when applying PeSTo to AlphaFold structures, concentrated in the flexible, lower-confidence regions where many interfaces occur; all reported relationships are associations, not demonstrated causes.

1 Introduction

Protein–protein interactions underlie nearly all cellular processes, from signal transduction to immune recognition, and the specific residues that form a binding interface are relevant to understanding disease mutations and to structure-based drug design [1]. Determining an interface experimentally, by solving the structure of a complex, remains slow relative to the scale of the problem: the Protein Data Bank (PDB) holds on the order of 2×10^5 structures, while the AlphaFold Protein Structure Database now provides predicted structures for over 2×10^8 sequences [2, 3]. Several geometric deep-learning methods now predict interface residues directly from a single protein’s three-dimensional coordinates, without requiring a solved complex: PeSTo represents atoms by element and position alone [4], and ScanNet extracts learned geometric and biochemical descriptors [5]. These models are trained on experimentally determined structures, and applying them to AlphaFold predictions implicitly assumes the predicted geometry is close enough to the true structure that the model’s learned decision boundary still applies.

That assumption is worth checking directly. AlphaFold reports a per-residue confidence score, pLDDT, and its documentation states that residues scoring below 70 may have unreliable backbone placement; low-confidence regions are disproportionately surface-exposed loops and chain termini, which is where many binding interfaces occur. Independent structural comparisons find that even confidently predicted regions can deviate from experimental structures by several tenths of an Ångström at longer range [6]. Yuan et al. benchmarked PeSTo and ScanNet on structures predicted by ESMFold, a protein-language-model-based structure predictor distinct from AlphaFold, and reported PeSTo’s AUPR falling from 0.797 on experimental structures to 0.691 on ESMFold structures [7, 8] – a substantial drop, but not a direct measurement of AlphaFold specifically, whose error characteristics differ from ESMFold’s.

This paper reports a matched, AlphaFold-specific benchmark of PeSTo together with an error analysis of where and why any accuracy gap occurs. We ask two questions: (1) how much does PeSTo’s interface-prediction accuracy change when the input is an AlphaFold model of a chain rather than that chain’s own experimental structure, holding the chain and the ground-truth interface fixed; and (2) is that change associated with AlphaFold’s own confidence score and with the local geometric divergence between the two structures. Both analyses were pre-registered: endpoints, statistical tests, and descriptive strata were fixed in writing before either comparison

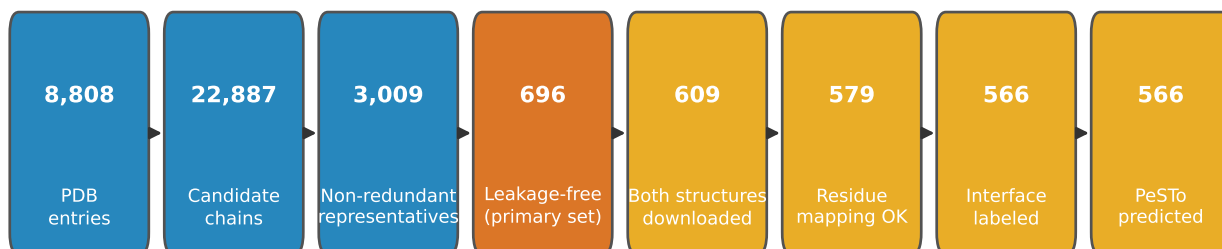


Figure 1: Dataset construction: candidate PDB chains are reduced by redundancy clustering, sequence-level leakage filtering against PeSTo’s own training/validation/test data, structure availability, residue-mapping validation, and interface labeling, to the 566-chain primary benchmark set. Counts are for the primary (leakage-filtered) track only.

was computed against real predictions, so that the reported result was not tuned after the fact. This extends prior experimental-to-predicted transferability work (including Yuan et al.’s ESMFold benchmark, and PeSTo’s own authors’ preliminary AlphaFold exploration) to a controlled, AlphaFold-specific comparison with a paired error analysis; it is not the first study to raise this general concern.

2 Methods

2.1 Dataset construction

Candidate protein chains were drawn from PDB entries released after 2018-04-30, AlphaFold2’s own training-data cutoff, so that the experimental structures used here could not have been memorized during AlphaFold’s training. Candidates were restricted to crystal structures at or better than 2.5 Å resolution, complexes of at most 10 distinct protein entities (excluding assemblies too large for a single well-defined pairwise interface, e.g. viral capsids), and chains of at least 40 modeled residues. This search returned 8,808 PDB entries and 22,887 candidate protein chains (Fig. 1). Redundant chains were then collapsed by MMseqs2 sequence clustering to 3,009 non-redundant representatives, one per cluster.

Because PeSTo was trained on essentially all biological assemblies in the PDB, filtering by exact PDB identifier is not sufficient: a chain released after 2018 can still be a close sequence homolog of something PeSTo has already seen under a different identifier. Each representative was therefore searched by MMseqs2 against the union of PeSTo’s published training, validation, and test sequence sets; 77% of representatives matched at or above 30% sequence identity to that union and were treated as leakage. The primary benchmark set used throughout this paper is the 696 representatives below this threshold.

For each primary-set representative, both the experimental (PDB) structure and the corresponding AlphaFold DB model were downloaded; 609 representatives had both available. AlphaFold DB occasionally serves third-party community structure models (e.g. ColabFold) under the same per-accession query as official AlphaFold2 predictions; these were identified from the cached prediction-API metadata’s provider field and excluded, since they do not carry AlphaFold2’s training-cutoff guarantee. Residue numbering between the experimental structure and the AlphaFold model (which uses UniProt-native numbering) was mapped residue by residue using each entry’s embedded SIFTS cross-references, falling back to a pairwise sequence alignment when SIFTS was unavailable or failed a $\geq 90\%$ identity validation check; 579 representatives obtained a validated mapping.

Ground-truth interface residues were computed directly from coordinates in the depositor-annotated biological assembly – not the raw crystallographic asymmetric unit, which is dominated by crystal-packing contacts unrelated to biological interfaces, and not the PISA assembly-annotation service [9], which has no practical batch interface for a set this size – as any residue within 5 Å of another protein chain. A second, buried-surface-area-based definition (loss of at least 1 Å² of solvent-accessible surface area upon complex formation) was retained as a robustness check on the labeling choice. 566 representatives received labels; this is the primary benchmark set analyzed in Section 3.

2.2 Inference

The pretrained PeSTo model (its element-only-input variant) was run on each of the 566 primary-set chains in two input forms: the isolated experimental chain (*exp*), and the AlphaFold model trimmed to the same mapped residue range (*af_trimmed*). Both runs used the chain in isolation, with no partner structure present, matching how an interface predictor would be applied to a single AlphaFold model in practice.

2.3 Pre-registered analysis plans

Two analyses were pre-registered in writing, with endpoints and thresholds fixed before either was run against real predictions.

Benchmark (Section 3.1). Primary and secondary endpoints are the paired per-chain difference (experimental minus AlphaFold-trimmed) in AUPR and ROC-AUC respectively, restricted to chains where both metrics are defined (excluding chains whose scored residues are single-class); tested with a two-sided Wilcoxon signed-rank test and a chain-resampled 95% bootstrap CI on the median difference (10,000 resamples, chains, not residues, resampled with replacement, since residues within one chain are not independent observations). Four descriptive strata (homomeric vs. heteromeric partner, small vs. large interface fraction, chains geometrically flagged by an independent AlphaFold-vs-experimental superposition check vs. not, all residues vs. surface residues only) and a robustness check against the buried-surface-area label were specified alongside the primary endpoint, not as additional hypothesis tests.

Error analysis (Section 3.2). pLDDT confidence bands are fixed at 50, 70, 90 (AlphaFold’s own convention), with pooled AUPR/ROC-AUC computed for both inputs within each band and a chain-resampled bootstrap CI (2,000 resamples per band, fewer than the benchmark’s 10,000 because each draw here recomputes a full pooled metric over tens of thousands of residues rather than a cheap scalar). A per-residue regression relates the absolute prediction shift, $|p_{AF} - p_{exp}|$, to standardized pLDDT, standardized local C α RMSD, standardized relative solvent accessibility (RSA), and secondary-structure class (coil as reference), with cluster-robust standard errors clustered by chain and variance inflation factors (VIF) reported for each predictor (flagged above 5, since pLDDT and local RMSD are both proxies for the same underlying AlphaFold-vs-experimental divergence and are expected to correlate). Local RMSD is computed with a separate rigid-body superposition per residue’s own 10 Å neighborhood, rather than one superposition for the whole chain, so that structural motion elsewhere in a multi-domain chain is not mistaken for local error at a given residue. Descriptive analyses of chain length and of the geometrically flagged chains were specified as associations to be reported, not confirmatory tests.

3 Results

3.1 Benchmark: experimental versus AlphaFold-trimmed input

Of the 566 primary-set chains, 12 (2.1%) were excluded from every paired comparison because every scored residue in the chain was labeled interface, leaving no non-interface residues against which to compute ROC-AUC or AUPR; the remaining 554 chains form the primary comparison.

PeSTo was more accurate on experimental input than on the matched AlphaFold-trimmed input: the median paired AUPR difference was 0.027 (95% CI [0.019, 0.035], Wilcoxon $p = 3.5 \times 10^{-17}$), and the paired ROC-AUC difference showed the same pattern (median 0.027, CI [0.018, 0.037], $p = 4.6 \times 10^{-19}$). Pooling all 95,118 scored residues (interface base rate 0.223) rather than comparing per-chain gives AUPR 0.739 on experimental input versus 0.669 on AlphaFold input (pooled ROC-AUC 0.882 vs. 0.849) – both inputs are far above the base rate, so both are informative, but the pooled AUPR gap (0.070) is noticeably larger than the per-chain median gap; Section 3.2 returns to why. Replacing the distance-based interface label with the buried-surface-area label reproduces essentially the same result (median 0.026, CI [0.020, 0.034], $p = 3.0 \times 10^{-17}$), so the finding is not an artifact of the distance-cutoff label definition.

All eight pre-registered strata showed the same direction of effect (Fig. 3), from a median 0.015 ($n=120$, Interface fraction ≥ 0.5) to 0.059 ($n=92$, Geometrically flagged); none reversed sign or lost significance in either direction. The gap is therefore not confined to one kind of chain, though its size varies noticeably – largest among the minority of chains flagged by an independent geometric check as showing unusual 3D disagreement between the AlphaFold model and the experimental structure.

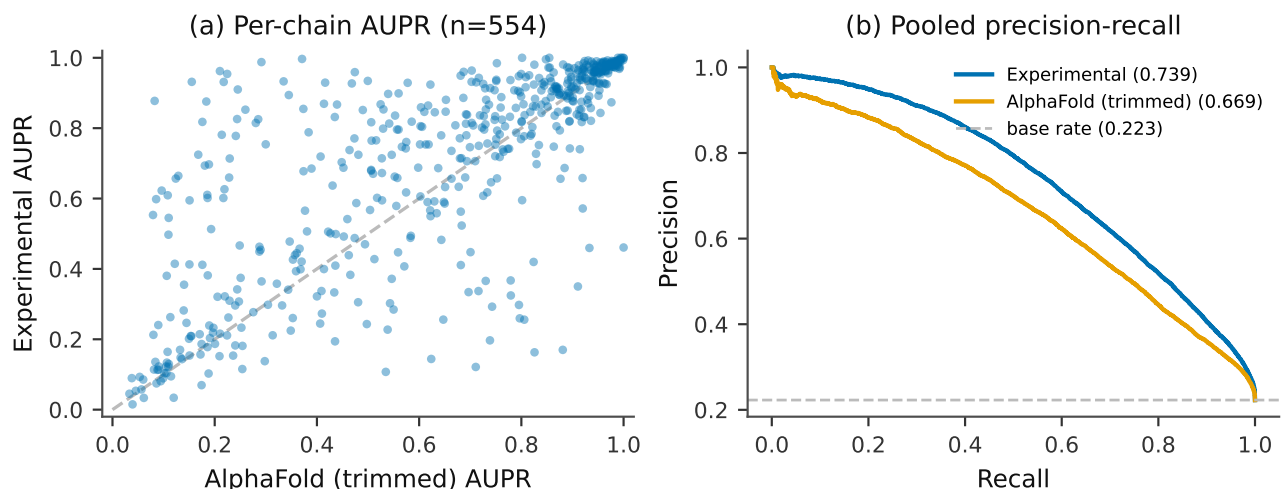


Figure 2: (a) Per-chain AUPR, experimental vs. AlphaFold-trimmed input; most chains sit near the identity line with a tilt toward experimental input, and a right-skewed tail where AlphaFold input costs noticeably more accuracy. (b) Precision–recall curves pooling every scored residue in the primary set, for each input, against the pooled interface base rate.

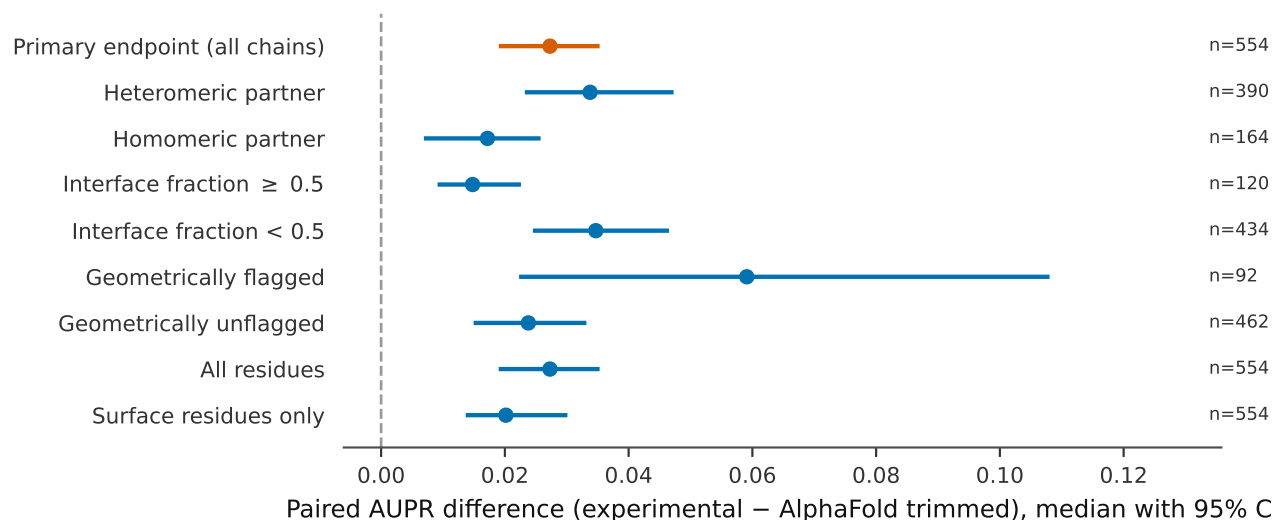


Figure 3: Paired AUPR difference (median, 95% chain-resampled bootstrap CI) for the primary endpoint and each of the eight pre-registered descriptive strata. All intervals lie on the same side of zero.

3.2 Error analysis: what the gap is associated with

Stratifying every scored residue by AlphaFold’s own pLDDT confidence band (Fig. 4, Table 1) shows the AUPR gap shrinking monotonically as confidence increases, from 0.254 in the lowest band to 0.039 in the highest – roughly a 6.4-fold reduction – with the same pattern in ROC-AUC. Every band cleared the minimum-residue reporting threshold with no undefined bootstrap draws.

A per-residue regression of the absolute prediction shift on standardized pLDDT, local RMSD, RSA, and secondary structure (Fig. 5, Table 2; $n = 95,117$ residues, 566 chain clusters, cluster-robust standard errors) finds pLDDT (coefficient -0.0268 , $p = 3.9 \times 10^{-16}$) and local RMSD (coefficient 0.0268 , $p = 1.4 \times 10^{-9}$) each independently associated with the size of the shift in the expected direction – higher confidence predicts a smaller shift, larger local structural divergence predicts a larger one – after controlling for the other and for RSA (coefficient 0.0254 , $p = 1.4 \times 10^{-57}$). No predictor’s variance inflation factor exceeded 5 (maximum 1.64): despite both being proxies for AlphaFold-vs-experimental divergence, pLDDT and local RMSD carry only mild collinearity here and each retains independent explanatory power. Consistent with the shift regression, mean absolute prediction shift rises steadily with local RMSD (Fig. 5a).

Two pre-registered descriptive analyses addressed possible confounds. First, per-chain AUPR drop showed no

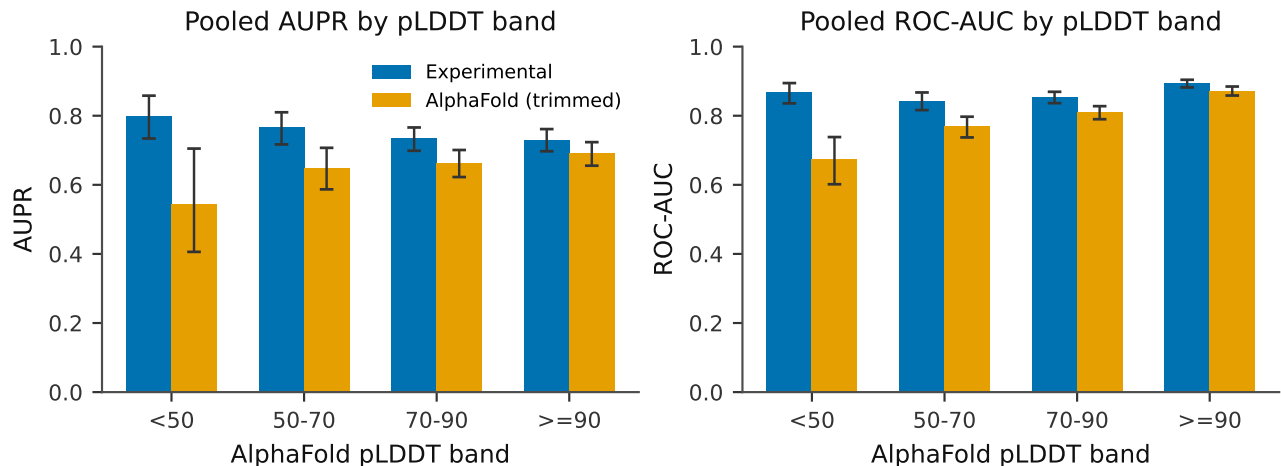


Figure 4: Pooled AUPR and ROC-AUC by AlphaFold pLDDT confidence band, both inputs, with 95% chain-resampled bootstrap CIs. The experimental-vs.-AlphaFold gap is largest in AlphaFold’s least confident residues and smallest in its most confident ones.

Band	<i>n</i> res.	<i>n</i> ch.	Base rate	Input	ROC-AUC (95% CI)	AUPR (95% CI)
<50	2,828	216	0.400	Experimental	0.868 [0.836, 0.894]	0.798 [0.734, 0.858]
				AlphaFold trimmed	0.673 [0.602, 0.738]	0.543 [0.406, 0.705]
50-70	5,735	445	0.332	Experimental	0.843 [0.816, 0.867]	0.766 [0.717, 0.810]
				AlphaFold trimmed	0.768 [0.737, 0.797]	0.649 [0.587, 0.707]
70-90	23,762	535	0.256	Experimental	0.853 [0.836, 0.869]	0.734 [0.699, 0.766]
				AlphaFold trimmed	0.809 [0.790, 0.828]	0.663 [0.623, 0.701]
>=90	62,793	493	0.192	Experimental	0.893 [0.882, 0.904]	0.730 [0.697, 0.761]
				AlphaFold trimmed	0.871 [0.858, 0.885]	0.690 [0.655, 0.724]

Table 1: Pooled ROC-AUC and AUPR by pLDDT band, both inputs, with 95% chain-resampled bootstrap CIs.

meaningful association with chain length (Spearman $\rho = 0.028$, 95% CI [-0.054, 0.108], $p = 0.51$) – this contradicts the pre-registration’s stated expectation that chain length would explain why the pooled AUPR gap (0.070) exceeds the per-chain median gap (0.027); the actual explanation is that the drop distribution is right-skewed (unweighted mean 0.061 vs. median 0.027), and weighting the mean by chain length barely moves it (0.061) – a minority of chains with an unusually large drop pull the mean well above the median, not long chains in general. Second, the 92 chains geometrically flagged for unusual 3D disagreement between AlphaFold and the experimental structure differ from the remaining 462 on every structural measure examined (Table 3): substantially lower mean pLDDT ($p = 2.8 \times 10^{-11}$) and higher local RMSD ($p = 4.2 \times 10^{-32}$), consistent with this flag capturing genuine AlphaFold-vs-experimental divergence rather than a mapping artifact.

Figure 6 illustrates these results on a single chain (PDB 7DZ9, chain C, UniProt E3BK13, 180 residues), chosen because its AUPR drop (0.031) sits close to the primary endpoint’s median. PeSTo’s experimental-input predictions (AUPR 0.837) track the true interface patch more tightly than its AlphaFold-input predictions (AUPR 0.806) on the same residues, though both visibly recover the same overall interface region.

4 Discussion

In practical terms, PeSTo’s accuracy cost from using an AlphaFold structure instead of an experimental one is real, directionally consistent, and statistically unambiguous, but modest for a typical chain: a median AUPR difference around 0.027 on a 0–1 scale is a small-to-moderate effect, not a collapse in accuracy, and Fig. 2a shows most chains clustering close to the identity line. The pooled-residue comparison (0.739 vs. 0.669) paints a superficially larger gap, but Section 3.2 shows that gap is driven by a right-skewed minority of chains rather than a uniform shift, so reporting the per-chain median alongside the pooled number matters for not overstating the typical case.

This project’s numbers are not directly comparable to Yuan et al.’s ESMFold-based benchmark (0.797 to 0.691 AUPR) [7]: the structure predictor, the candidate set, the leakage control, and the interface definition all differ,

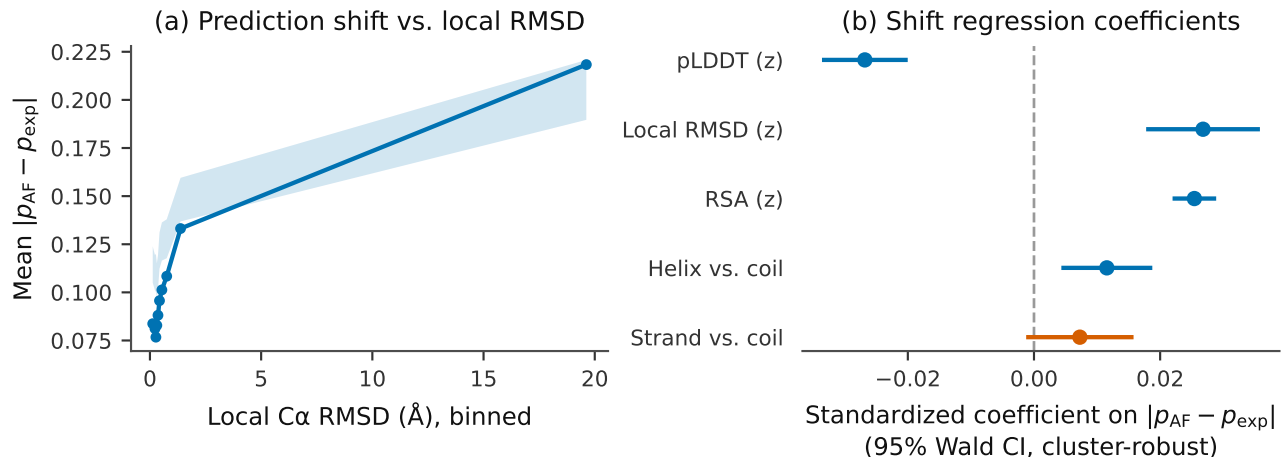


Figure 5: (a) Mean absolute prediction shift by binned local C α RMSD, with chain-resampled 95% CIs. (b) Standardized coefficients (95% Wald CI) from the per-residue shift regression; grey markers did not reach $p < 0.05$.

Term	Coef.	Cluster-robust SE	t	p	VIF
Intercept	0.1019	0.0029	35.17	6.8×10^{-271}	–
pLDDT (z)	-0.0268	0.0033	-8.14	3.9×10^{-16}	1.64
Local RMSD (z)	0.0268	0.0044	6.05	1.4×10^{-9}	1.48
RSA (z)	0.0254	0.0016	15.99	1.4×10^{-57}	1.18
Helix vs. coil	0.0115	0.0035	3.30	9.8×10^{-4}	1.15
Strand vs. coil	0.0073	0.0042	1.74	0.081	1.18

Table 2: Regression of $|p_{AF} - p_{exp}|$ on standardized predictors, cluster-robust standard errors by chain. Reference category: secondary structure = coil.

and only the direction of the effect – a real accuracy cost from using a predicted structure – is expected to generalize between the two studies, not the magnitude. AlphaFold’s average accuracy is generally reported as higher than ESMFold’s [8], which is at least directionally consistent with this study’s smaller median per-chain gap, but the two benchmarks differ in enough other respects (dataset, leakage definition, exact interface criterion) that this is a plausible reading of the comparison, not a controlled one.

The error analysis in Section 3.2 points to a specific, mechanistic account of the gap rather than a diffuse "AlphaFold structures are worse" story: the accuracy cost concentrates in residues where AlphaFold itself reports low confidence and where the local backbone geometry genuinely diverges from the experimental structure, and these two signals carry largely independent information (low collinearity despite both tracking the same underlying divergence). That pLDDT alone predicts where PeSTo will be less reliable on AlphaFold input – without needing the experimental structure to compute local RMSD, which is of course unavailable at deployment time – is the most actionable finding here: it suggests a practical, currently unused confidence signal for flagging low-reliability regions of PeSTo’s predictions on AlphaFold-only inputs.

5 Limitations

- **Temporal skew.** The primary set’s leakage-filtering requirement (sequence identity below 30% against PeSTo’s training data) leaves it skewed toward chains released in 2021 or later, farther in time from AlphaFold’s 2018-04-30 cutoff than the filter alone requires; PeSTo’s own training-data snapshot appears to extend to roughly 2020–2021, later than AlphaFold2’s cutoff, based on a release-year breakdown of the leakage flag (not shown here).
- **Bound vs. unbound conformations.** Both experimental structures here are *bound*, complexed structures; a protein’s shape can change somewhat upon binding, and this study cannot separate that ordinary conformational change from AlphaFold-specific structural error, because the independently solved unbound (apo) arm designed to make that separation was not run in this pass.
- **Deferred comparisons.** The untrimmed, full-length AlphaFold model (as opposed to the same model trimmed to the experimentally observed range) and the leakage-sensitivity subsets at looser identity thresholds were not

Metric	Flagged mean (median)	Unflagged mean (median)	p
Per-chain AUPR drop	0.108 (0.0591)	0.052 (0.0238)	0.020
Chain length (residues)	224 (202)	162 (117)	2.5×10^{-4}
Interface fraction	0.314 (0.215)	0.327 (0.266)	0.179
Mean pLDDT	77.4 (84.1)	87.9 (90.6)	2.8×10^{-11}
Mean local RMSD (Å)	2.21 (1.39)	0.672 (0.55)	4.2×10^{-32}
Mean global C α dist. (Å)	7.91 (7.37)	1.4 (0.98)	9.5×10^{-40}

Table 3: Geometrically flagged vs. unflagged primary-set chains (two-sided Mann-Whitney U test).

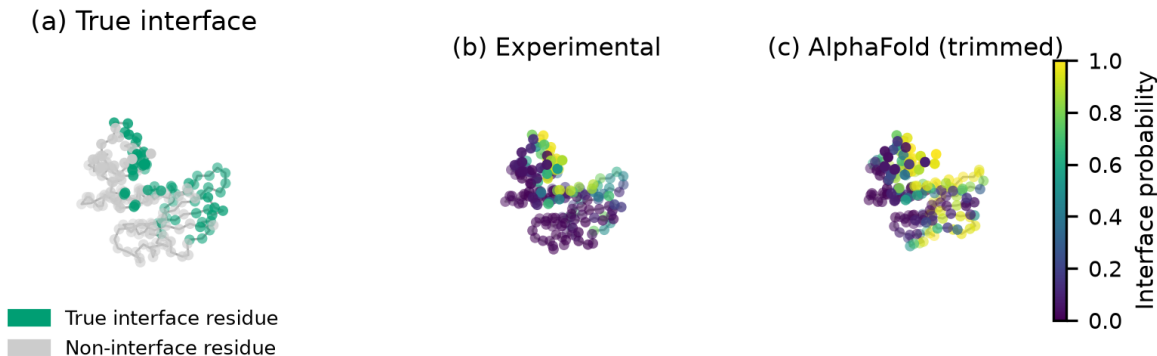


Figure 6: A representative chain (C α trace; no headless 3D molecular renderer could be installed in this environment, so this is a matplotlib rendering of C α positions colored by value, not a rendered surface – see Methods discussion in the accompanying repository documentation). (a) True interface residues (green) vs. non-interface (grey). (b) PeSTo’s per-residue interface probability from the experimental structure (AUPR 0.837 on this chain). (c) PeSTo’s per-residue interface probability from the AlphaFold model trimmed to the same residues (AUPR 0.806), superposed onto the experimental frame for comparison only (the underlying coordinates, and hence the prediction, come from the AlphaFold model).

run in this pass; the trimmed-model result reported here should not be read as characterizing performance on a full-length AlphaFold model used without any trimming.

- **Single predictor, single confidence signal.** All results are specific to PeSTo; whether they generalize to ScanNet [5], HIGH-PPI [10], or other interface predictors is untested here. Only pLDDT was used as AlphaFold’s confidence signal; predicted aligned error (PAE), which is specific to multi-chain predictions, was not available since every AlphaFold input here is a single trimmed monomer model.
- **Monomeric AlphaFold models.** Every AlphaFold structure used here is a single-chain model; no multi-chain (complex) AlphaFold prediction was evaluated, so this study speaks to isolated-chain structure quality, not to whether an AlphaFold-predicted complex’s own interface would look different.

6 Future Work

The deferred arms noted above – the untrimmed full-length AlphaFold comparison, the unbound/apo arm, and the leakage-sensitivity subsets at looser identity thresholds – remain natural next steps and require no new method, only additional runs of the existing pipeline. Given that pLDDT is independently associated with prediction reliability (Section 3.2), a pLDDT-aware fine-tuning of PeSTo – adding pLDDT as an extra per-residue input channel, training on a mixed set of experimental structures (pLDDT fixed at 100) and AlphaFold structures (their real pLDDT) – is a plausible next step toward recovering some of the measured accuracy cost, conditional on this gap and its pLDDT association, both established here.

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